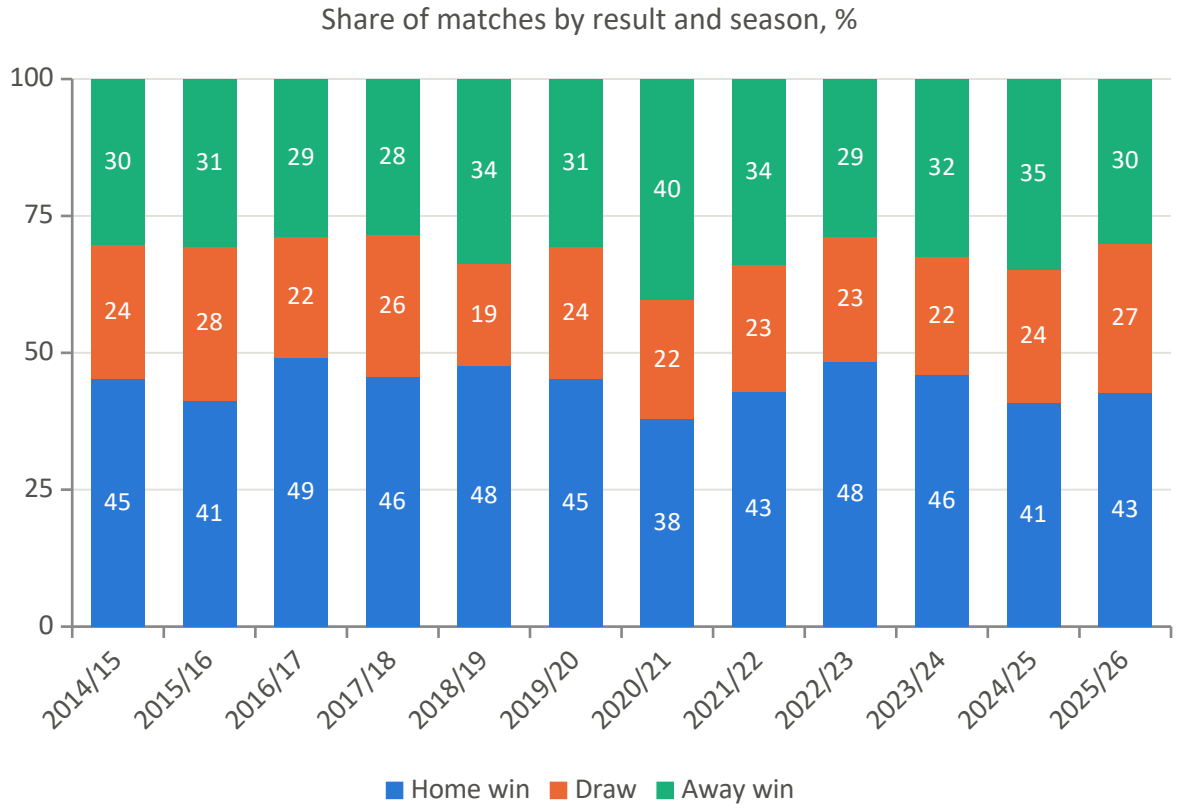
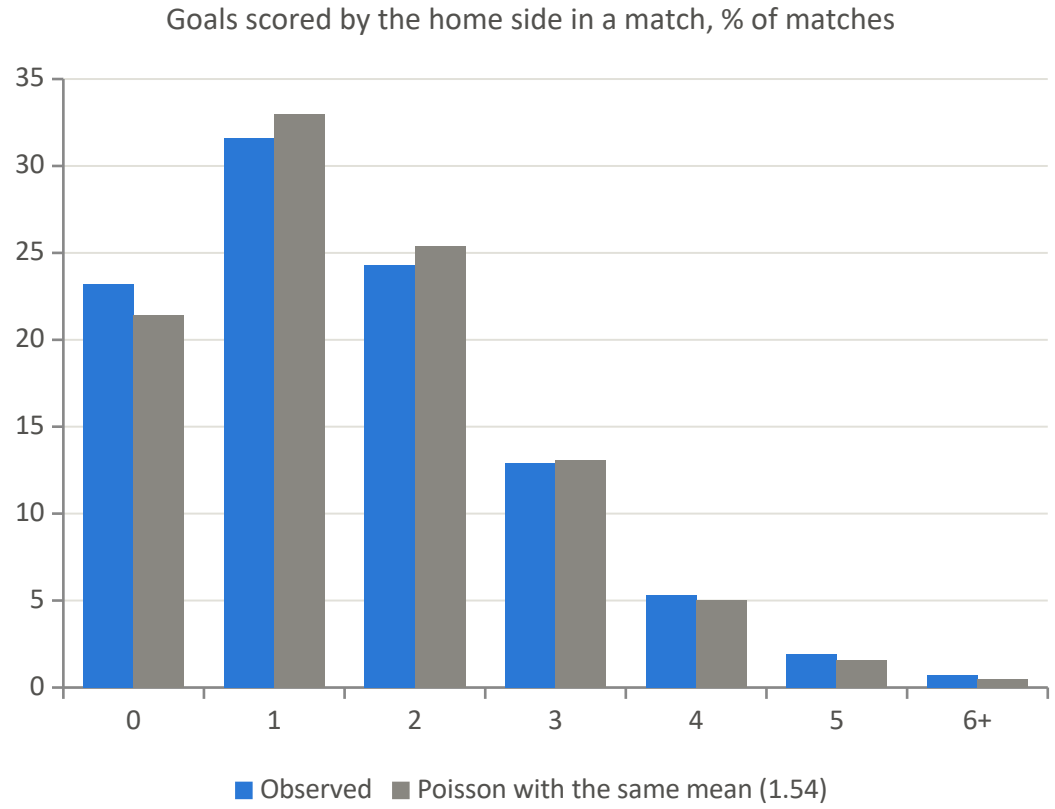
Predicting Premier League match results

Project phase 2: exploratory analysis, models, error analysis

Home sides win more; goals follow Poisson

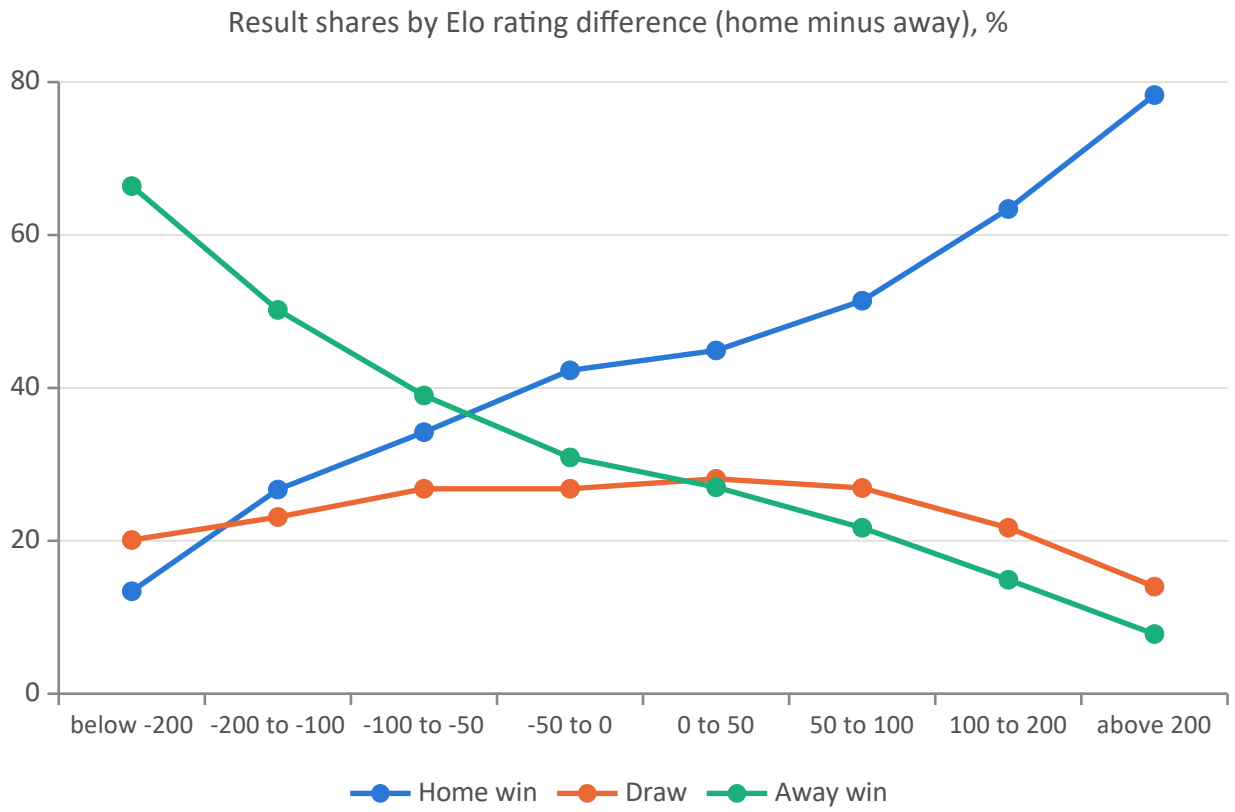


Over 4,560 matches: home 44%, draw 24%, away 32%. Lowest home share: 38% in 2020/21, played mostly without crowds.

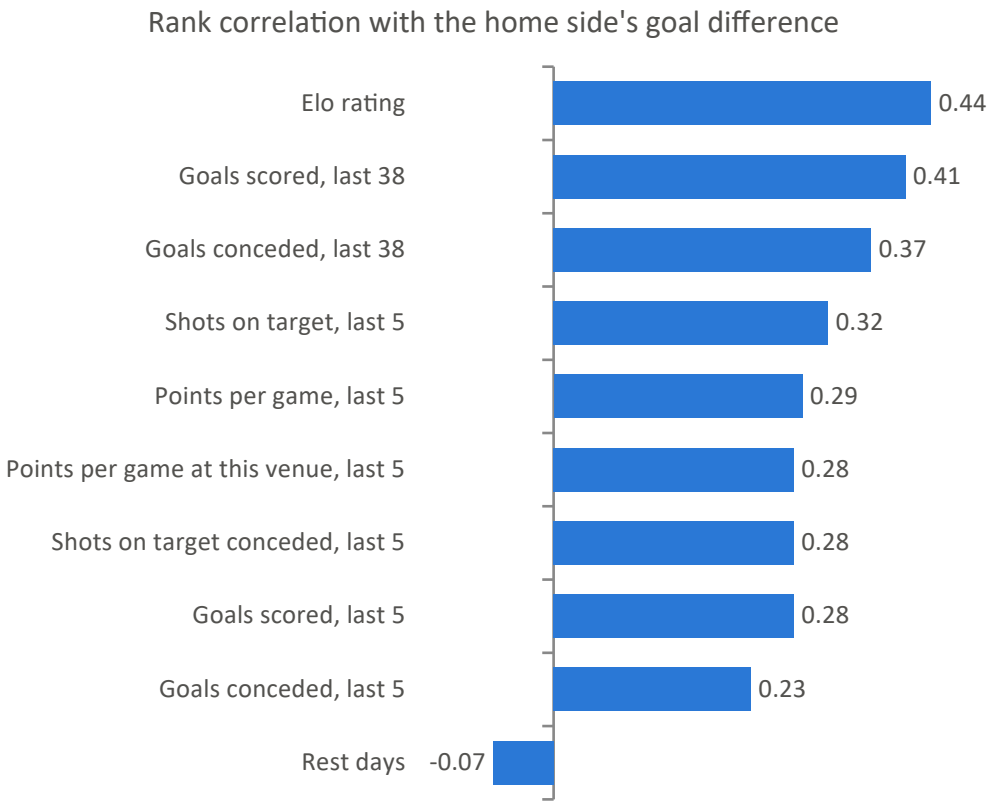


Every observed share is within 1.9 points of the Poisson value. The goal model rests on this.

Long-run strength predicts results; form adds little



With a rating lead above 200 the home side wins 78% of matches. Draws never exceed 28% in any group.



Each bar is the home side's advantage on one feature. Ratings and season-long goals lead; five-match form trails; rest days carry nothing.

Six models, one walk-forward protocol

Class frequencies

The same three probabilities for every match. The floor.

Elo + logistic regression

One feature: the rating difference.

Poisson goal model

Attack and defence strength per club give expected goals and a score grid.

Logistic regression

All 21 features, multinomial, regularised.

Random forest

400 trees on the 21 features.

Gradient boosting

Shallow boosted trees on the 21 features.

Twelve seasons, used in order

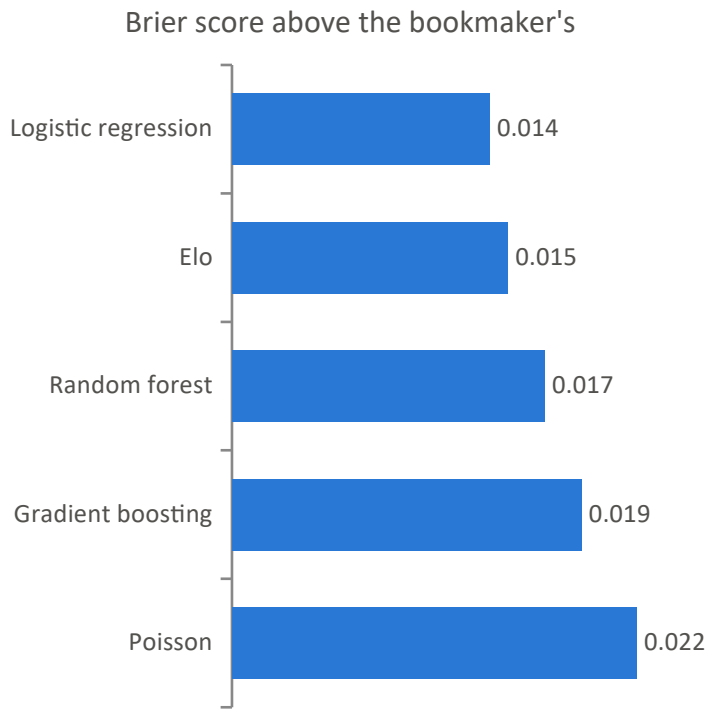


- **Warm-up** 2014/15
- **Training** 2015/16 to 2020/21
- **Validation** 2021/22, 2022/23
- **Test** 2023/24 to 2025/26

- Settings chosen on the validation seasons, then frozen
- Before each test season a model is refitted on every earlier season; the Poisson model before every match day
- 1,140 test matches, scored by accuracy, Brier score and log loss
- Reference: Bet365's odds at kickoff, margin removed

No model reaches the bookmaker

Model	Accuracy	Brier	Log loss	Brier gap to bookmaker, 95% interval
Logistic regression	**53.9%**	**0.584**	**0.980**	**+0.014 (+0.007 to +0.021)**
Elo + logistic regression	53.4%	0.585	0.981	+0.015 (+0.008 to +0.022)
Random forest	53.6%	0.587	0.986	+0.017 (+0.010 to +0.023)
Gradient boosting	52.5%	0.589	0.990	+0.019 (+0.011 to +0.026)
Poisson goal model	52.0%	0.592	0.994	+0.022 (+0.013 to +0.031)
Class frequencies	43.2%	0.650	1.073	+0.080 (+0.063 to +0.095)
Bookmaker (Bet365 at kickoff)	**55.0%**	**0.570**	**0.960**	**reference**



- Best model: logistic regression, Brier 0.584 against the bookmaker's 0.570; every interval excludes zero
- The rating difference alone (0.585) matches all 21 features; tree models are slightly worse
- All models are far better than class frequencies (0.650)

Draws are a quarter of results and never predicted

Logistic regression

Actual	Pred. home	Pred. draw	Pred. away
Home	410 83%	0 0%	82 17%
Draw	178 64%	0 0%	101 36%
Away	164 44%	0 0%	205 56%

Bookmaker

Actual	Pred. home	Pred. draw	Pred. away
Home	403 82%	0 0%	89 18%
Draw	171 61%	0 0%	108 39%
Away	145 39%	0 0%	224 61%

Rows: what happened. Columns: the result each forecaster rated most likely. Cells: matches, and share of the row.

0 of 279

draws predicted by the model, and by the bookmaker

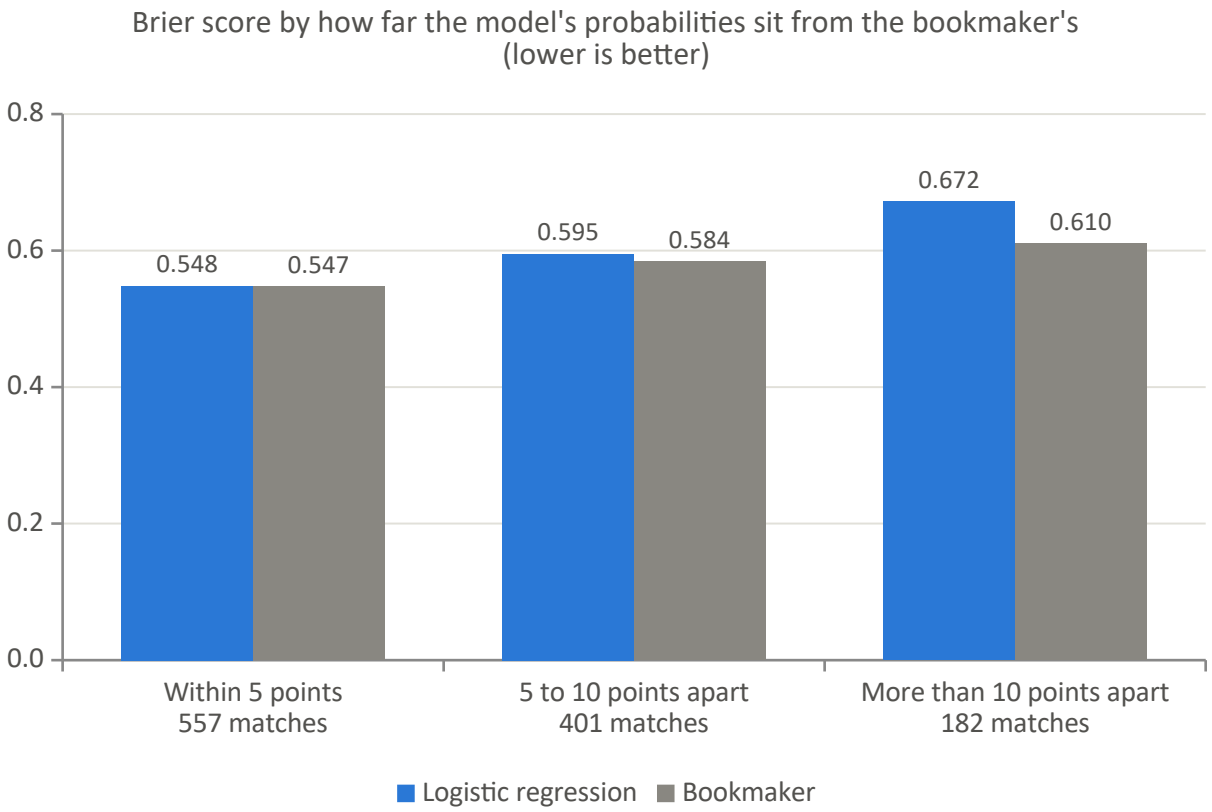
30%

is the highest draw probability the model ever gives

2.5 pts

largest gap between predicted probability and observed frequency

When they disagree, the bookmaker is right



In 557 matches the two agree within 5 points and score the same. In the 182 where they differ by more than 10, the bookmaker is ahead by 0.062.

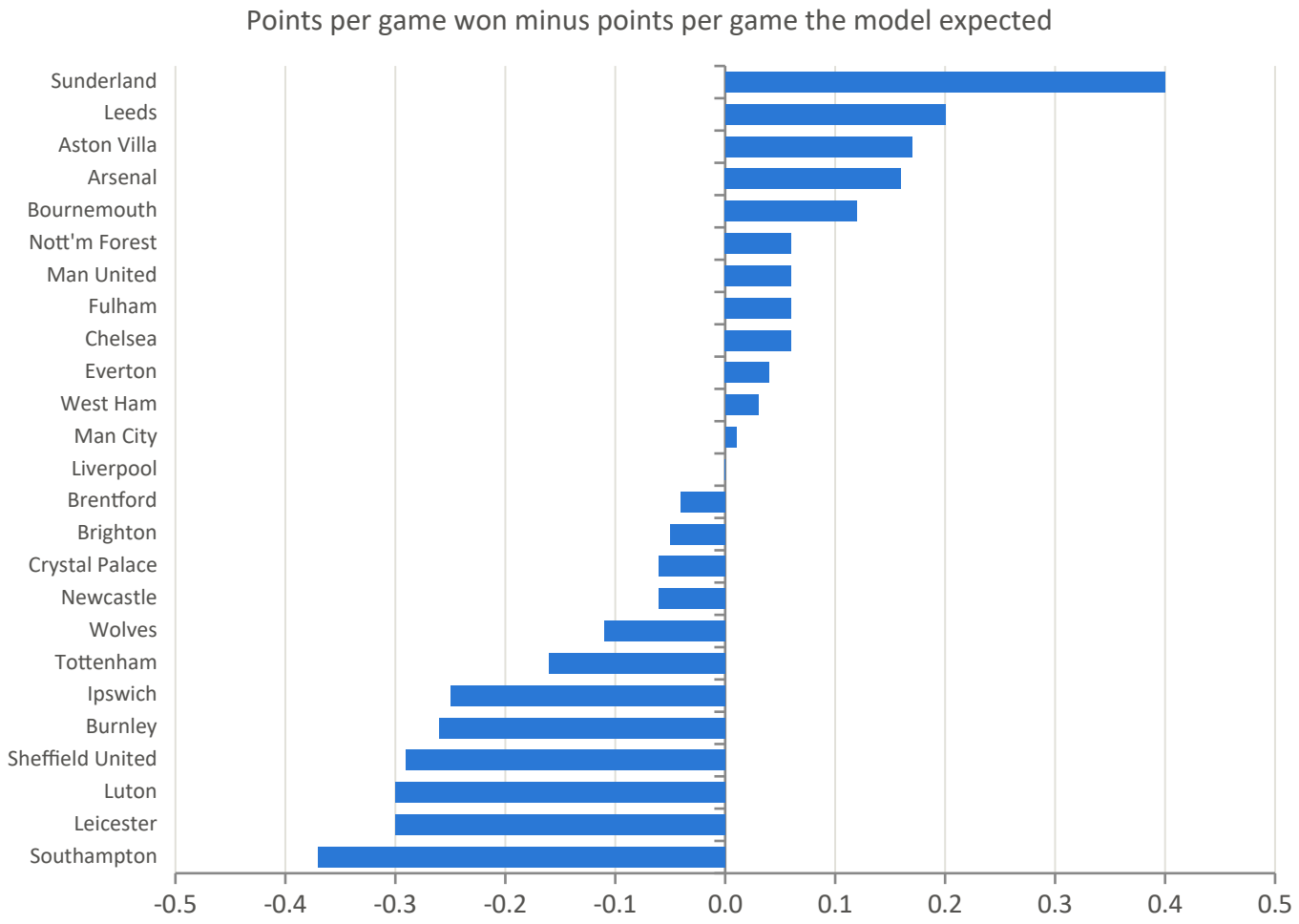
Betting on the disagreement

One unit on every result a model rates above +5% expected value at the kickoff odds.

Model	Bets	Won	Return
Elo	724	30%	**−7.0%**
Logistic regression	740	27%	**−9.0%**
Gradient boosting	811	26%	**−15.1%**
Random forest	767	25%	**−17.9%**
Poisson	872	26%	**−18.5%**

Every model loses money. A disagreement with a stronger forecaster is mostly the weaker one's error.

Clubs new to the league are the blind spot



8 largest errors

are all clubs that spent only one or two of the three seasons in the league. A promoted club arrives with no history, so the model gives it an average newcomer's rating.

Upsets surprise everyone

The result the model thought least likely: Liverpool 0-1 Crystal Palace, 6% for the model and 9% for the bookmaker. That part of the error is chance.

What we learned

A simple, calibrated model gets most of the way

Logistic regression: 54% accuracy, Brier 0.584 against 0.650 for class frequencies.

The bookmaker stays ahead

Brier 0.570. Its odds reflect line-ups, injuries and transfers that results alone do not show.

More complexity did not help

One rating feature matches 21 features, and tree ensembles are slightly worse on a few thousand training matches.

Next steps

Squad and line-up data, a model aimed at draws, and better starting ratings for promoted clubs.